

Ananth Sriram

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science + Machine Learning

College Park, MD

May 2026

EXPERIENCE

Capital One

Associate Software Engineer

Aug 2026 – Present

McLean, VA

- Selected for Capital One's Technology Development Program (TDP).

Capital One

Software Engineering Intern

Jun 2025 – Aug 2025

Richmond, VA

- Productionized a zero-touch issue-triaging pipeline on **AWS Lambda** using **Llama 4 Scout**, replacing a legacy SVM classifier and its 3-hour manual CSV workflow with automated processing in **10 minutes**.
- Orchestrated a scheduled **Jira-to-model retraining loop** that shifted the bot from passive ticket-filing to autonomous resolution, closing **78%** of tickets without manual intervention.
- Created a containerized Selenium module to programmatically execute data uploads via **AWS ECS Fargate** and **Docker**, diagnosing and resolving **10+ critical Docker runtime failures** related to memory constraints and architecture mismatches.
- Automated credential rotation for **8** service accounts via **AWS Secrets Manager**, closing a SOC2 audit finding and eliminating manual credential handling for admin-interface access.

Columbia University - CRIS Lab

Undergraduate Research Assistant

Sep 2025 – Present

New York City, NY

- Engineered an **online reinforcement learning** algorithm using the **Multi-Armed Bandit (RL)** approach for a **genetic algorithm framework**, dynamically balancing operator allocation to achieve a **27.3%** improvement in success rate in ground-truth mechanism recovery.
- Co-authored and selected for an **oral presentation** at the *American Institute of Chemical Engineers Annual Meeting* (**Second Author**) for engineering a **novel framework** for symbolic regression.

University of Maryland - Precision Agriculture Lab

Undergraduate Research Intern

Dec 2023 – Present

College Park, MD

- Built a **Google Earth Engine** and **Python** pipeline integrating decades of **open-source** geospatial and tabular data (Sentinel-2, NASS); the resulting framework is under revision at the *Journal of Applied Earth Observation and Geoinformation* (**First Author**).
- Co-authored a century-scale **predictive modeling** study coupling NA-CORDEX climate projections with FAO-56 water-balance modeling to quantify corn yield stability under non-adaptive irrigation strategies in Maryland (**Second Author**), under revision at *Agricultural Water Management*.
- Co-authored "An Overview of Drones in Agriculture" (**Second Author**), survey paper for Machine Learning applications regarding drone-based precision agriculture.

PROJECTS

1st Place, \$10K Winner | *Ironsight.ai (South Park Commons, 8VC) Spatial Intelligence Hackathon, UMD* Feb 2026

- Built a 3-stage ML pipeline (**YOLO11, SAM3, Qwen3-VL-8B**) to detect **PPE non-compliance** and proximity hazards from egocentric construction footage, mapping violations to **OSHA** standards with severity scoring; work **published** as an **arXiv** preprint (**First Author, endorsed by Dr. Ramani Duraiswami**).
- Designed a 3-pass adversarial chain-of-thought **VLM** protocol with structurally enforced pass independence, achieving a **12% precision improvement**.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Go, SQL, Ruby

Frameworks/Data: React, Flask, Apache Spark, Kafka, Postgres, ETL Pipelines

Cloud/Dev Tools: AWS (ECS Fargate, Lambda, ALB, SNS), Docker, Linux, Git, Cursor, Claude Code, pytest

AI/ML Tools: PyTorch, TensorFlow, scikit-learn, AutoML, MLOps, Pandas, NumPy